Architecture Blueprint

STY1004 Capstone Project — Phase 1

Project	Downtown Sudbury traffic-light IoT platform (Paris Street corridor / Bridge of Nations)
Course	STY1004 — Software Development Capstone
Team	Ryan Spencer, Dylan Brost, Zach Swain, Alex Schroeder
Repository	https://github.com/ryanspencer20/STY1004---Capstone-Project
Date	6 November 2026

1. System Overview

1.1 Problem

This capstone is based on a sequential corridor of traffic lights in downtown Sudbury, Ontario. Consecutive intersections on the Paris Street corridor sit close together. When their timings are not coordinated, vehicles hit red lights in a row during peak hours. The City of Greater Sudbury has been working to fix uncoordinated corridors of this kind (Clarke, *Sudbury.com*, 15 August 2023).

The platform is meant to augment that infrastructure with simulated camera telemetry: vehicle counts for smarter green time, and red-light / speed monitoring for infractions. Physical bulbs stay with the existing PLC; this software models intersection state, rules, camera data, and suggested phase changes.

1.2 Platform scope

The solution is a C# / .NET clean-architecture IoT platform. A rules engine combines fixed timing (min/max green, yellow, advanced green) with count-based overrides when an approach is crowded. Simulated camera payloads stand in for live video. Violation logic logs red-light and speeding events (plate and speed when available). EF Core persists locations, devices, signal state, rules, telemetry, and tickets so the team can inspect history and tune thresholds.

Identity uses serial numbers, not numeric IDs: **INT-###** is the place (the intersection stays if a camera is swapped); **CAM-###** is the camera hardware at that place. Example corridor points: INT-001 Paris & Cedar, INT-002 Paris & John, INT-003 Paris toward Van Horne / Bridge of Nations.

1.3 Solution layout

- **IoTPlatform.Domain** — entities, value objects, enums, invariants. No database or UI references.
- **IoTPlatform.Application** — application services (device actions, intersection/corridor actions, violation alerts).
- **IoTPlatform.Infrastructure** — EF Core persistence (SQLite in the current repo build), Fluent API mapping, migrations, and repositories.
- **IoTPlatform.UnitTests** — MSTest coverage of domain rules and application services.

1.4 In scope / out of scope

In scope: simulated camera metrics, signal-state rules and hard safety invariants (a North–South green forces East–West red and Don't Walk, and the reverse; no green↔red skip of yellow), timed schedules plus count overrides, historical storage, console UI, isolated unit tests.

Out of scope: live video decode, ML training, vehicle-type recognition, installing hardware on city poles, emergency-vehicle or transit GPS preemption.

2. Domain Model Diagram

Class diagram of Phase 1 domain types: value objects and enumerations (gray) and entities (blue). Lights and timing rules belong to the location. Counts and tickets belong to the camera device. Persistence may store current lights in a separate **Signal_States** table; in the domain those fields still live on **IntersectionLocation**.

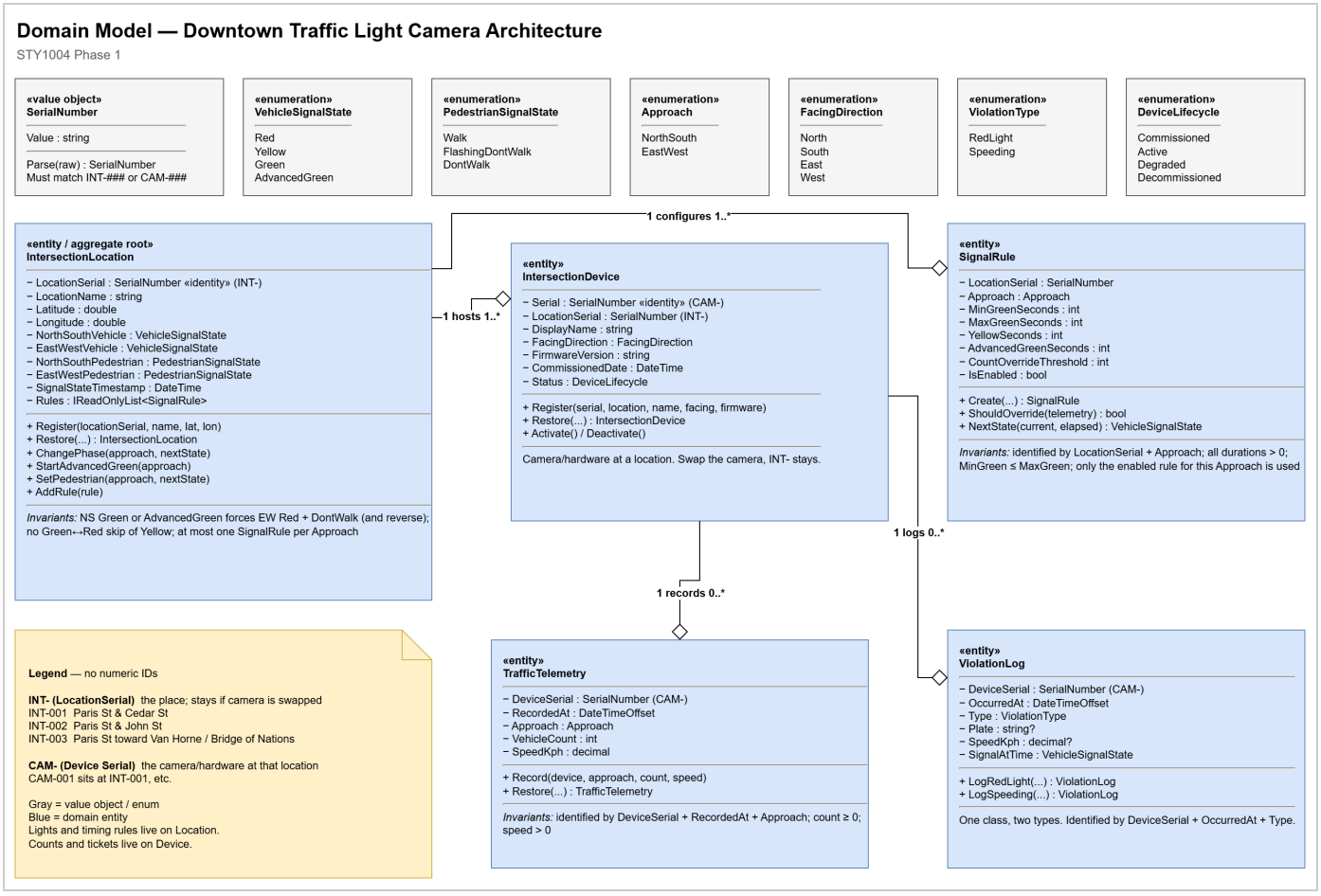


Figure 1. Domain model — downtown traffic light camera architecture. Source: `Phase 1/Domain-Model-Diagram.png` on `main`.

Relationships: one **IntersectionLocation** hosts one or more **IntersectionDevice** cameras and configures one or more **SignalRule** rows (one enabled rule per approach). Each device records zero or more **TrafficTelemetry** samples and zero or more **ViolationLog** tickets (red light or speeding).

SerialNumber must match `INT-###` or `CAM-###`. Location methods enforce opposite-approach lock (green / advanced green vs red + Don't Walk) and refuse skipping yellow. **SignalRule.ShouldOverride** / **NextState** decide when counts justify a phase change. The program advises the PLC; it does not drive the bulbs directly.

3. Entity Relationship Diagram (ERD)

Relational schema for keys, constraints, and data types, mapped with EF Core Fluent API and generated by the InitialCreate migration. Six tables:

- **Locations** — place identity (location serial PK), name, latitude, longitude.
- **IntersectionDevices** — camera (device serial PK), foreign key to location.
- **Signal_States** — current NS/EW vehicle and pedestrian lights plus timestamp; one row per location.
- **Signal_Rules** — timings and count-override threshold; composite key location serial + approach.
- **Traffic_Telemetries** — vehicle count and speed; composite key device serial + recorded-at + approach.
- **Violation_logs** — red-light / speeding tickets; composite key device serial + occurred-at.

Rules and signal state belong to the location. Telemetry and violations belong to the device.

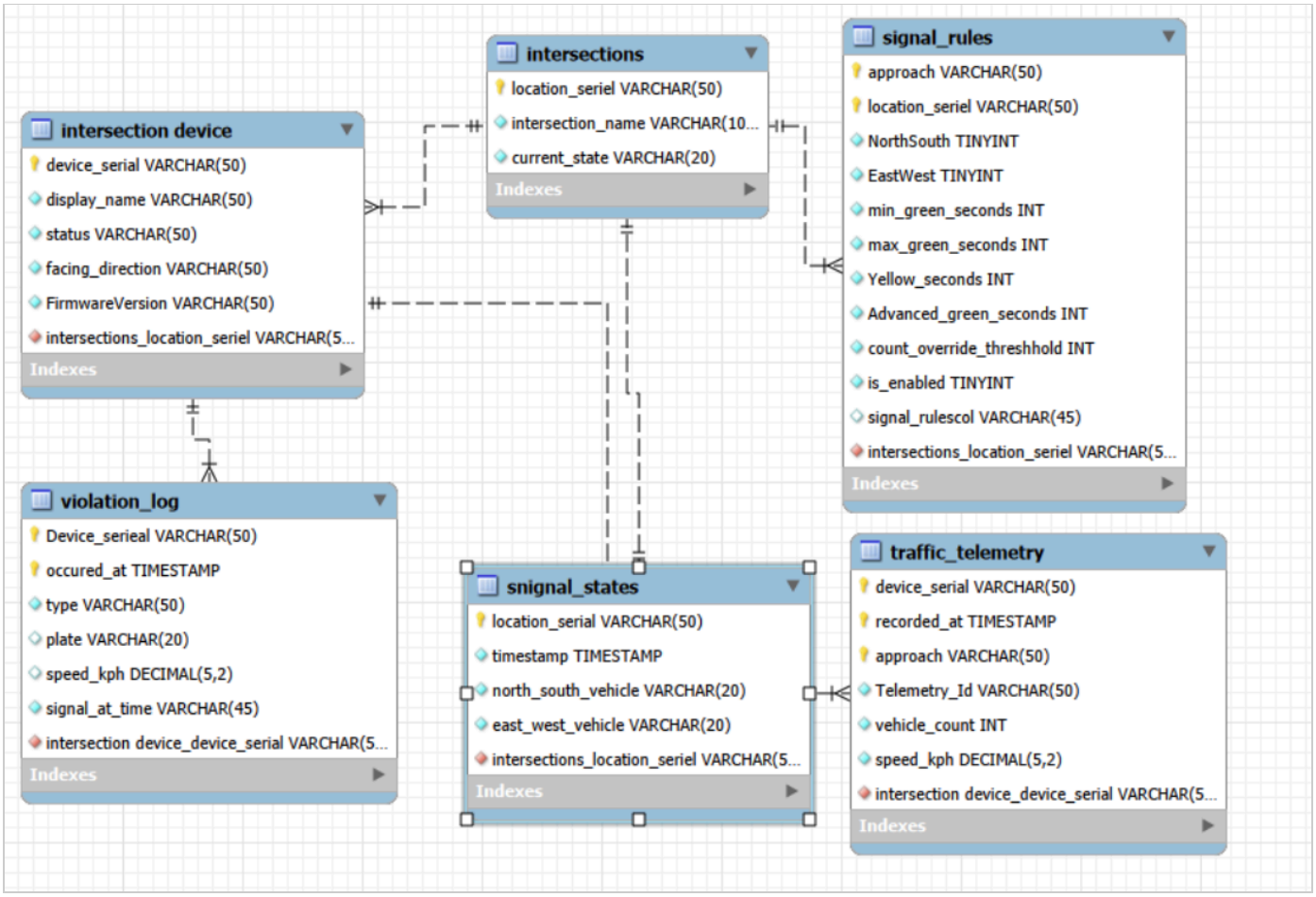


Figure 2. Entity relationship diagram. Source: ERD on `main` (merged from the Phase 1 ERD pull request).

4. Team Contributions

Phase 1 responsibilities and git commits on ryanspencer20/STY1004---Capstone-Project (`main`).

Team member	GitHub	Responsibilities (Phase 1)	Git commits
Ryan Spencer	ryanspencer20	Repo setup and README / project scope; EF Core database, migrations, and repositories; Application layer (intersection / device / violation services); merge integration on `main`.	`4d06368` Update README Project Scope `f2f2581` Database EFCore initial build `1981c69` Expand Domain and Database for corridor override `c166e2d` Finalize Application layer, database, and unit testing `6ddacc2` Merge pull request #17 (EF Core database)
Dylan Brost	DylanBrost	Four-project solution scaffold; domain classes (serial numbers, telemetry, signal rules); MSTest coverage of domain rules and later combined tests.	`32383de` Set up four-project solution and references for #5 `b0afaea` Serial numbers, telemetry validation, enums, and tests `acd0a93` Remaining domain classes, signal rules, and unit tests `4a3ed86` Cleaned tests to one-act / DataRow `5e9cf9b` Merge Ryan's code with latest tests
Zach Swain	zachAswain	Phase 1 domain class diagram (entities, value objects, relationships). Updated from Guid IDs to INT-/CAM-serials and the location vs camera split.	`55703f5` Add Phase 1 domain class diagram (PR #10) `fc53cac` Replace Guid IDs with unique serial numbers and device names (PR #13) `017e514` Update Phase 1 domain class diagram (PR #18)
Alex Schroeder	Cenedrine	Entity relationship diagram: six tables, keys, and relationships for locations, devices, signal state, rules, telemetry, and violations.	`a63d895` Add files via upload (ERD screenshot) `209e1cf` Merge pull request #14 (ERD diagram)

5. Reference

[1] T. Clarke, “Hate red light waits? Sudbury project optimizing traffic flow,” *Sudbury.com*, 15 August 2023. https://www.sudbury.com/local-news/hate-red-light-waits-sudbury-project-optimizing-traffic-flow-7408291